IS101 — Introduction to Computing

Inventory & Budget Workbook

Oasis Events · Unit 3 — Spreadsheets · Variation 2 · 100 points · 25 tasks

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How to use this document: Download the starter workbook (Excel_Var2_Starter.xlsx) from the Project Grader page. Open it in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order. The Inventory sheet has 8 items (rows 4–11) with columns for Item ID, Item Name, Category, Qty, Unit Cost, Line Total, SKU, Category Code, and Item Label. The PriceList sheet tab has item names and unit costs for lookups. When you're done, save the file and upload it to the Variation 2 grader for instant scoring. **Important:** Task 1 turns on Show Formulas mode, which makes every formula display as text instead of a calculated value. You can toggle this on and off at any time with Ctrl+` to check your work. When writing formulas, always reference cell addresses, not column name words.

Workbook Setup Tasks 1–4

Task 1 2 points

Turn on Show Formulas

WHY THIS MATTERS

Show Formulas mode displays every formula in its cell instead of the result — useful for auditing. **Note:** With Show Formulas on, every formula you enter will display as text instead of its result. To verify your formulas are working, press Ctrl+` to toggle Show Formulas off, check your results, then press Ctrl+` again to turn it back on.

WHERE TO FIND IT

Formulas tab → Formula Auditing group → Show Formulas (or press Ctrl+`).

HOW YOU'LL KNOW IT WORKED

Every cell shows its formula text instead of calculated values. Column widths widen automatically.

Task 2 3 points

Set workbook properties: Title and Author

WHY THIS MATTERS

Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT

File tab → Info → Properties panel. Set Title to “Event Inventory & Budget” and Author to your name.

HOW YOU'LL KNOW IT WORKED

File → Info shows Title and Author populated.

Task 3 3 points

Use SEQUENCE to generate the line-item ID column

WHY THIS MATTERS

SEQUENCE creates a series of sequential numbers automatically — useful any time you need auto-numbered rows without typing them by hand. **Important:** enter it as one formula per row, not one formula that fills all 8 rows at once. A single =SEQUENCE(8) in A4 tries to spill its results down into A5:A11 automatically — but once column A becomes part of an Excel Table (Task 5) with a Total Row (Task 7), Excel no longer allows a spilling formula to expand inside a Table. It will show a #VALUE! error and stop updating. Entering one non-spilling SEQUENCE formula per row avoids this completely, and still uses the SEQUENCE function as required.

WHERE TO FIND IT

Click cell A4 → type **=SEQUENCE(1,1,ROW()-3,1)** → Enter. This returns a single number (1) for row 4, not an array. Copy that formula down through A11 (drag the fill handle, or copy/paste) — because ROW() changes on each row, every copy calculates its own correct ID automatically (2, 3, 4 … 8).

HOW YOU'LL KNOW IT WORKED

Toggle Show Formulas off (Ctrl+`) to verify: column A reads 1 through 8, one number per row, with no error values.

Task 4 4 points

Format Unit Cost and Line Total columns as Currency

WHY THIS MATTERS

Currency formatting adds dollar signs and two decimal places to financial data.

WHERE TO FIND IT

Select the Unit Cost column data (E4:E11) and the Line Total column data (F4:F11) → Home tab → Number group → Currency.

HOW YOU'LL KNOW IT WORKED

Values show as $12.00 format with dollar signs and two decimals.

Tables & Data Tasks 5–9

Task 5 3 points

Convert the inventory data to an Excel Table

WHY THIS MATTERS

Excel Tables add automatic filtering, banded rows, and structured references. **Important:** only convert the core inventory columns — Item ID through Item Label (A3:I11) — into the Table. Do not extend it into the Event Date, Vendor, or summary columns further to the right (columns K onward); those need to stay as regular cells so the SORT and UNIQUE formulas in Tasks 14–15 can spill freely. (Excel Tables never allow a formula's results to spill across rows — the same reason Task 3 uses a per-row SEQUENCE.)

WHERE TO FIND IT

Click any cell inside A3:I11 → Ctrl+T (or Insert tab → Table, or Home tab → Format as Table). In the Create Table dialog, confirm the range shown is =$A$3:$I$11 — not a wider range — then click OK.

HOW YOU'LL KNOW IT WORKED

Filter arrows appear on columns A through I only (not on Event Date, Vendor, or the summary columns further right); the Table Design tab appears when you click inside the table.

Task 6 4 points

Confirm the table covers every item and the Line Total column

WHY THIS MATTERS

The grader checks that your table's range actually covers all 8 inventory items (rows 4–11) plus the Line Total column — not a partial selection. If the range gets shrunk while creating the table (for example, only grabbing a few rows, or stopping before column F), later tasks like the Total Row (Task 7) won't have Line Total to work with.

WHERE TO FIND IT

Click inside your table → Table Design tab → Resize Table (or drag the small resize handle in the table's bottom-right corner) → confirm the range is A3:I11 (headers in row 3, all 8 items in rows 4–11, columns A through I).

HOW YOU'LL KNOW IT WORKED

Scrolling through the table shows all 8 items with no gaps, and the Line Total column (F) is included with the rest of the table's banded formatting.

Task 7 4 points

Add a Total Row summing the Line Total column

WHY THIS MATTERS

Complete Task 8 (Line Total formula) and Task 13 (XLOOKUP for Unit Cost) first so there are values to sum. **Important:** turning on the Total Row checkbox by itself is not enough. Excel automatically puts a Sum in whichever column happens to be rightmost in your table (in this workbook that's Item Label, not Line Total) — so if you check the box and stop there, the total lands under the wrong column and this task is marked incorrect even though a Total Row exists. You have to manually tell Excel which column to sum.

WHERE TO FIND IT

Click anywhere in the table → Table Design tab → check the Total Row checkbox (a row appears at the bottom with Sum already applied to the rightmost column). Click the Total Row cell directly under Line Total (column F) → click the small dropdown arrow that appears → choose Sum from the list.

HOW YOU'LL KNOW IT WORKED

The Total Row cell under Line Total (F) — not any other column — shows the sum of all 8 line totals as a dollar amount.

Task 8 6 points

Calculate the Line Total

WHY THIS MATTERS

The Line Total column (F) multiplies each item's Qty by its Unit Cost. Complete Task 13 (XLOOKUP) first so the Unit Cost column has values, then come back to this task.

WHERE TO FIND IT

Click cell F4 → type a formula that multiplies the Qty in D4 by the Unit Cost in E4 → Enter. Copy the formula down for all rows (F5 through F11).

HOW YOU'LL KNOW IT WORKED

Toggle Show Formulas off to verify: each row shows Qty × Unit Cost (e.g., 120 × $3.50 = $420.00).

Task 9 6 points

Use SUMIF to total spend per category

WHY THIS MATTERS

SUMIF adds values that match a condition. The category names are pre-filled in column O (O4:O9). Write SUMIF formulas in column P to calculate the total spend for each category.

WHERE TO FIND IT

Click cell P4 → type a SUMIF formula that searches the Category column (C4:C11) for the category name in O4, and sums the corresponding Line Total values (F4:F11) → Enter. Copy down through P9.

HOW YOU'LL KNOW IT WORKED

Each category in column O shows its total spend in column P (e.g., Furniture = $600.00).

Text & Lookup Formulas Tasks 10–16

Task 10 4 points

Use LEFT, RIGHT, or MID to build a short SKU

WHY THIS MATTERS

Text functions extract parts of a string. Build a SKU code in column G by combining characters from the Item Name (B) and/or Category (C). Example: first 3 letters of name + “-” + first 2 letters of category = “FOL-FU” for Folding Chair/Furniture.

WHERE TO FIND IT

Click cell G4 → type a formula using LEFT and/or MID to extract characters from columns B and C → Enter. Copy down for all rows.

HOW YOU'LL KNOW IT WORKED

Column G shows short codes (at least 3 characters each) for every item.

Task 11 4 points

Use UPPER and LEN for a category code

WHY THIS MATTERS

UPPER converts text to all capitals; LEN counts characters. Combine them to create a category code in column H.

WHERE TO FIND IT

Click cell H4 → type a formula that combines UPPER of the Category (C4) with the LEN of the Category → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column H shows codes like “FURNITURE9” — the uppercase category name followed by its character count.

Task 12 5 points

Use TEXTJOIN to build a combined item label

WHY THIS MATTERS

TEXTJOIN concatenates multiple values with a delimiter between them. Build a label in column I combining Item Name, Category, and Qty using a dash as the delimiter.

WHERE TO FIND IT

Click cell I4 → type a TEXTJOIN formula using “ - ” (dash with spaces) as the delimiter, skipping empty cells, and joining the Item Name (B4), Category (C4), and Qty (D4) → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column I shows labels like “Folding Chair - Furniture - 120” for each item.

Task 13 6 points

Use XLOOKUP to return each item’s unit cost from the PriceList

WHY THIS MATTERS

XLOOKUP searches for a value and returns a corresponding result from another column. Look up each Item Name (column B) on the PriceList sheet to fill in the Unit Cost (column E). Do this task before Task 8 so the Line Total formula has values to multiply.

WHERE TO FIND IT

Click cell E4 → type an XLOOKUP formula that searches for the Item Name (B4) in the PriceList sheet’s item name column and returns the matching unit cost → Enter. Copy down for all rows.

HOW YOU'LL KNOW IT WORKED

Column E shows the correct unit cost for each item (e.g., Folding Chair = $3.50).

Task 14 4 points

Use SORT() to produce a sorted vendor list

WHY THIS MATTERS

The SORT function creates a dynamically sorted list. Place it in the Sorted Vendors area (column Q). Because column Q is outside your Table (see Task 5), this formula can spill freely down into Q4:Q8 with no conflict.

WHERE TO FIND IT

Click cell Q4 → type a SORT formula referencing the Vendor column (M4:M8) → Enter.

HOW YOU'LL KNOW IT WORKED

Column Q shows the vendor names sorted alphabetically.

Task 15 4 points

Use UNIQUE() to list distinct categories

WHY THIS MATTERS

The UNIQUE function removes duplicates from a list. Place it in the Categories area (column R). Because column R is outside your Table, this formula can spill freely down into R4:R9 with no conflict.

WHERE TO FIND IT

Click cell R4 → type a UNIQUE formula referencing the Category column (C4:C11) → Enter.

HOW YOU'LL KNOW IT WORKED

Column R shows each category name once, with no duplicates.

Task 16 5 points

Use YEAR, MONTH, and DAY to break out the event date

WHY THIS MATTERS

Date functions extract parts of a date. The event date is in cell K4. Break it into Year, Month, and Day.

WHERE TO FIND IT

Click cell T4 → type =YEAR(K4) → Enter. Click U4 → type =MONTH(K4) → Enter. Click V4 → type =DAY(K4) → Enter.

HOW YOU'LL KNOW IT WORKED

T4 shows 2026, U4 shows 9, V4 shows 15.

Conditional Formatting & Named Range Tasks 17–18

Task 17 5 points

Apply an icon set to flag items at reorder level

WHY THIS MATTERS

Icon sets add visual indicators (arrows, flags, or traffic lights) based on cell values.

WHERE TO FIND IT

Select the Qty column data (D4:D11) → Home tab → Conditional Formatting → Icon Sets → choose a style.

HOW YOU'LL KNOW IT WORKED

Each cell in the Qty column shows a colored icon indicating the relative quantity level.

Task 18 4 points

Name the price-list range “PriceList”

WHY THIS MATTERS

Named ranges make formulas easier to read.

WHERE TO FIND IT

Click the PriceList sheet tab → select the price data range (A4:B11) → click in the Name Box → type PriceList → press Enter.

HOW YOU'LL KNOW IT WORKED

The Name Box shows “PriceList” when the range is selected.

Charts Tasks 19–21

Task 19 4 points

Create a Pie chart of spend by category

WHY THIS MATTERS

A Pie chart shows how total spend is divided among categories.

WHERE TO FIND IT

Select the category names and spend totals in the SUMIF summary area (O4:P9) → Insert tab → Charts group → Pie → choose a style.

HOW YOU'LL KNOW IT WORKED

A pie chart appears showing each category’s share of total spend.

Task 20 4 points

Apply a chart style

WHY THIS MATTERS

Chart styles change the overall look of the chart.

WHERE TO FIND IT

Click the chart → Chart Design tab → Chart Styles gallery → choose a style.

HOW YOU'LL KNOW IT WORKED

The chart’s appearance changes to the selected style.

Task 21 3 points

Add alt text to the chart

WHY THIS MATTERS

Alt text describes the chart for screen readers and accessibility.

WHERE TO FIND IT

Right-click the chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED

Right-clicking the chart and selecting Edit Alt Text shows your description.

Finishing Touches Tasks 22–25

Task 22 4 points

Add a header with the company name and a footer with page number

WHY THIS MATTERS

Headers and footers identify printed pages. **Important:** Use the Page Setup method below, not Insert → Header & Footer, which switches views and can unfreeze your panes from Task 24.

WHERE TO FIND IT

Page Layout tab → click the Page Setup dialog launcher (small arrow in the bottom-right corner) → Header/Footer tab → Custom Header: type “Oasis Events” → Custom Footer: insert page number → OK.

HOW YOU'LL KNOW IT WORKED

Print Preview shows the company name at the top and page number at the bottom.

Task 23 3 points

Merge and center the title banner

WHY THIS MATTERS

Merging creates a wide banner that spans the data columns.

WHERE TO FIND IT

Select cells A1 through the last data column in row 1 → Home tab → Alignment group → Merge & Center.

HOW YOU'LL KNOW IT WORKED

The title “Oasis Events — Inventory & Budget” spans all data columns in one centered cell.

Task 24 3 points

Freeze the header row

WHY THIS MATTERS

Freezing keeps the column headers visible while scrolling down through inventory items.

WHERE TO FIND IT

Click cell A4 (the first data row) → View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

When you scroll down, row 3 (the headers) stays visible at the top.

Task 25 3 points

Sort by Category then Item Name

WHY THIS MATTERS

Multi-level sorting groups items by category and alphabetizes within each group.

WHERE TO FIND IT

Data tab → Sort → Sort by: Category (column C) → A to Z → Add Level → Then by: Item Name (column B) → A to Z → OK.

HOW YOU'LL KNOW IT WORKED

Items are grouped by category; within each category, items are alphabetical.

Total: 100 points. Every task is graded automatically when you upload your file. © Dr. Mark Taormino · All Rights Reserved · College of Southern Nevada